2.3 Forced Response to Nonperiodic Excitations:

- **Solution is transient; no steady-state**
- **Convolution integral methods (semi-analytical solutions)**
- **Transform methods (e.g., Laplace transform approach)**

2.3.1 Response to specific inputs:

- **Impulsive response function**

$$\int_{-\infty}^{\infty}\delta(t-a)dt=1\,,\qquad F(t)=\hat{F}\delta(t-a)$$

$$m\ddot{g}(t)+c\dot{g}(t)+kg(t)=\delta(t)$$

$$g(t)=\begin{cases}\dfrac{1}{m\omega_d}e^{-\zeta\omega_n t}\sin\omega_d t\,, & t>0\\ 0\,, & t<0\end{cases}$$

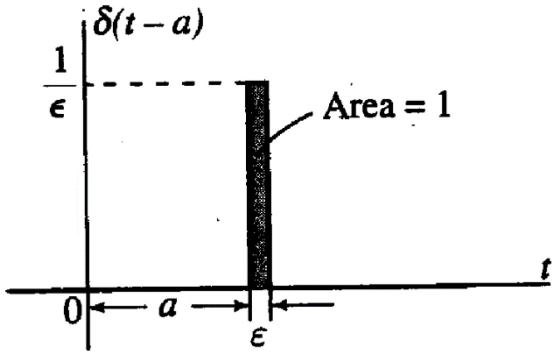


Fig. 2.12 Unit impulse or Dirac-delta function.

- **Unit step response function**

$$\boldsymbol{u}(t-a)=\begin{cases}0, & t<a\\ 1, & t>a\end{cases}\;;\quad \boldsymbol{u}(t-a)=\int_{-\infty}^{t}\delta(\tau-a)\,d\tau$$

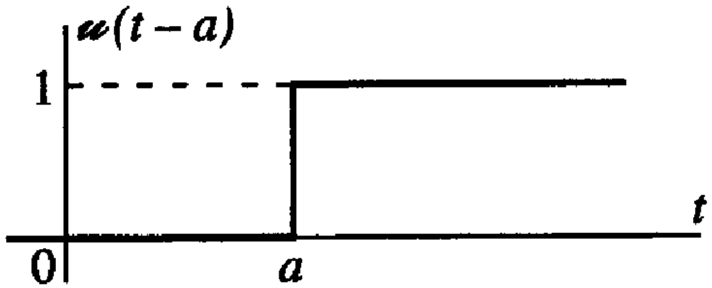
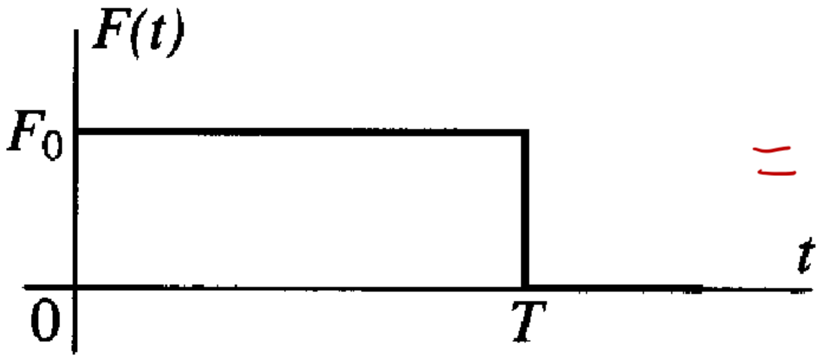


Fig. 2.13 Unit step function.

$$\boldsymbol{s}(t)=\int_{-\infty}^{t}g(\tau)\,d\tau=\frac{1}{k}\left[1-e^{-\zeta\omega_n t}(\cos\omega_d t+(\zeta\omega_n/\omega_d)\sin\omega_d t)\right]\boldsymbol{u}(t)$$

Example (Text Example 4.1):

$$m\ddot{x}(t)+kx(t)=F(t)\,,\quad F(t)=F_0\left[\boldsymbol{u}(t)-\boldsymbol{u}(t-T)\right]$$

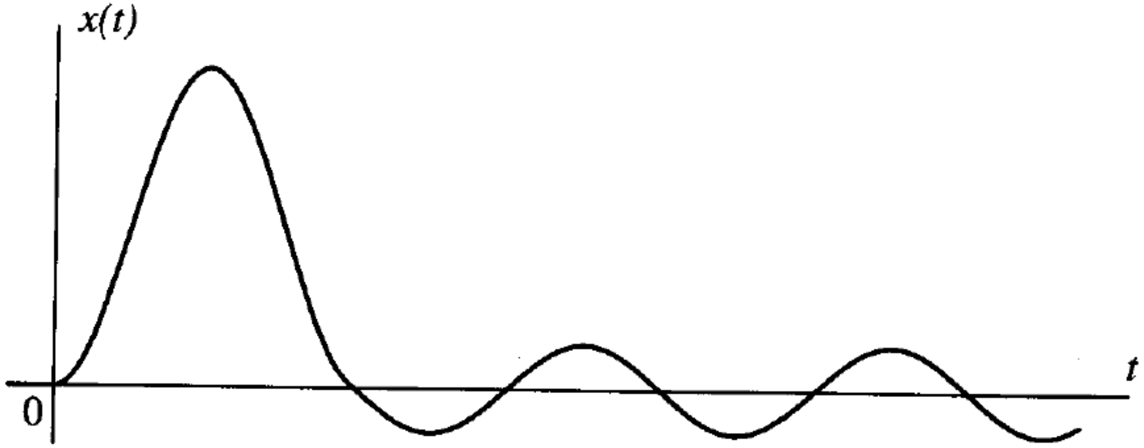


=

minus

$$x(t)=F_0\left[\boldsymbol{s}(t)-\boldsymbol{s}(t-T)\right]$$

$$=\frac{F_0}{k}\left\{(1-\cos\omega_n t)\boldsymbol{u}(t)-[1-\cos\omega_n(t-T)]\,\boldsymbol{u}(t-T)\right\}$$



2.3.1 Response to specific inputs (continue):

- **Unit ramp response function**

$$r(t-a)=(t-a)\boldsymbol{u}(t-a)=\int_{-\infty}^{t}\boldsymbol{u}(\tau-a)\,d\tau$$

$$\boldsymbol{u}(t-a)=\frac{d}{dt}(r(t-a))$$

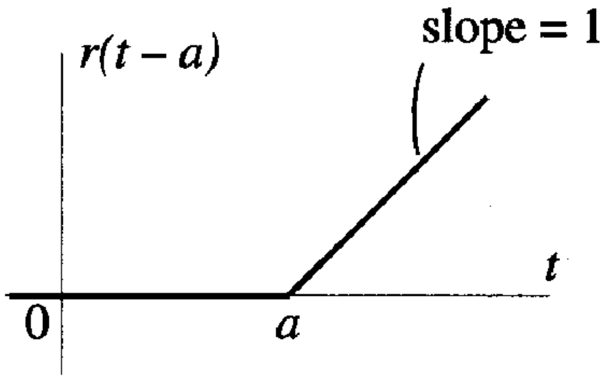
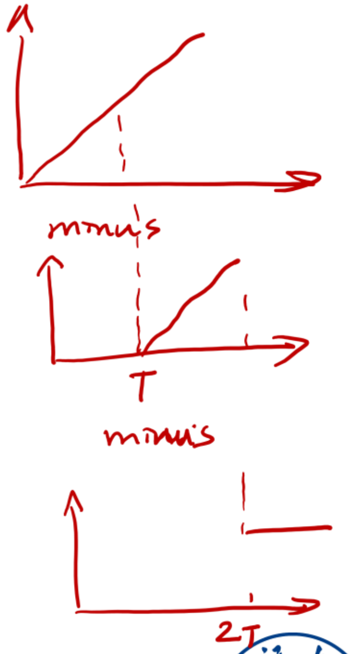


Fig. 2.14 Unit ramp function.

$$(m\frac{d^2}{dt^2}+c\frac{d}{dt}+k)\,\mathbf{r}(t)=r(t)$$ with zero initial conditions

$$\int_{-\infty}^{t}(m\frac{d^2}{dt^2}+c\frac{d}{dt}+k)\boldsymbol{s}(\tau)d\tau=\int_{-\infty}^{t}\boldsymbol{u}(\tau)d\tau \quad\Rightarrow\quad \mathbf{r}(t)=\int_{-\infty}^{t}\boldsymbol{s}(\tau)d\tau=\int_{0}^{t}\boldsymbol{s}(\tau)d\tau$$

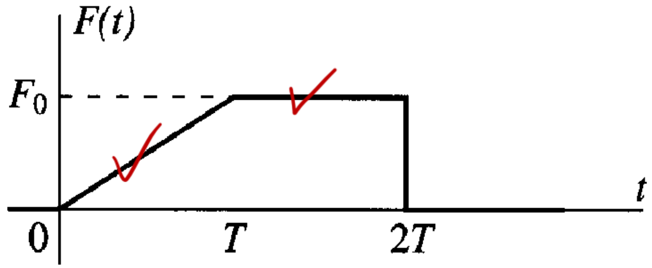
$$\mathbf{r}(t)=\frac{1}{k}\left[t-\frac{2\zeta}{\omega_n}+e^{-\zeta\omega_n t}\left(\frac{2\zeta}{\omega_n}\cos\omega_d t+\frac{2\zeta^2-1}{\omega_d}\sin\omega_d t\right)\right]\boldsymbol{u}(t)$$



Example:

$$F(t)=F_0\left[\frac{1}{T}r(t)-\frac{1}{T}r(t-T)-\boldsymbol{u}(t-2T)\right]$$

$$x(t)=F_0\left[\frac{1}{T}\mathbf{r}(t)-\frac{1}{T}\mathbf{r}(t-T)-\boldsymbol{s}(t-2T)\right]$$



2.3.2 Response to arbitrary excitation by convolution integral:

- Represent the shaded area as an impulse with strength $\hat{F}(\tau)$

$$\hat{F}(\tau)\delta(t-\tau) = F(\tau)\Delta\tau\ \delta(t-\tau)$$

$$\Rightarrow F(t) \cong \sum F(\tau)\Delta\tau\ \delta(t-\tau)$$

Dirac-delta fn

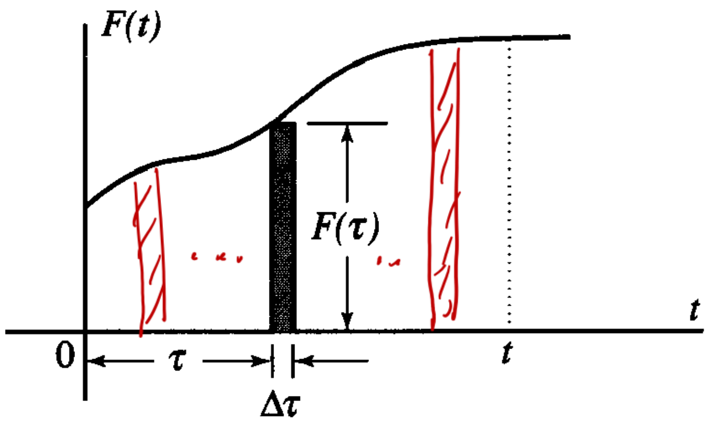


Fig. 2.15 Arbitrary excitation as an convolution integral.

- Regard F(t) as a linear superposition of impulsive forces and the response is written in terms of delayed impulse response function

$$x(t) = \sum F(\tau)\Delta\tau\ g(t-\tau)$$

As $\Delta\tau \to 0,$ $$\boxed{x(t) = \int_0^t F(\tau)g(t-\tau)\,d\tau = \int_0^t F(t-\lambda)g(\lambda)\,d\lambda}$$

- The convolution integral is thus symmetric. Numerical and analytical evaluations usually choose the simpler of the two functions in the argument in the shifting (time delay) process.

- Consider F(t) as an unit step function: $x(t) = \int_0^t u(t-\tau)g(\tau)\,d\tau = \int_0^t g(\tau)\,d\tau$

Duhamel's integral formulation:

- An alternate approach to the convolution integral approach
- Contribution of the response of a step function of amplitude ΔF(τ) is:

$$\Delta x \cong \Delta F(\tau)\,\mathbf{s}(t-\tau) = \frac{\Delta F(\tau)}{\Delta \tau}\,\mathbf{s}(t-\tau)\,\Delta\tau$$

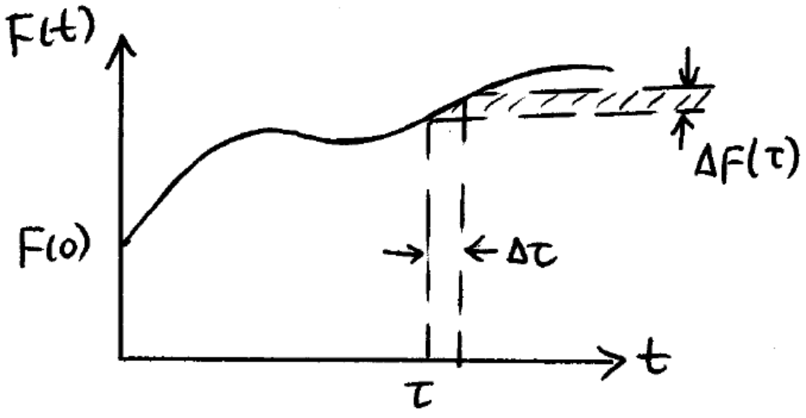


Fig. 2.16 Arbitrary excitation as a sum of step functions.

As $\Delta\tau \to 0,\ x(t) = F(0)\mathbf{s}(t) + \int_0^t \frac{dF(\tau)}{d\tau}\,\mathbf{s}(t-\tau)\,d\tau$

Apply integration by parts: $x(t) = \mathbf{s}(0)F(t) + \int_0^t \frac{d\mathbf{s}(t-\tau)}{d\tau}\,F(\tau)\,d\tau$

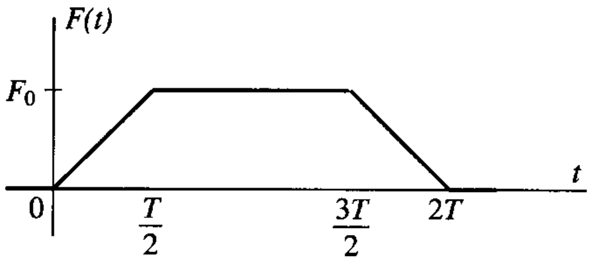
$$\boxed{x(t) = \frac{d}{dt}\int_0^t \mathbf{s}(t-\tau)\,F(\tau)\,d\tau}$$

Substituting $\mathbf{s}'(t-\tau) = g(t-\tau) - \mathbf{s}(0)\,\delta(t-\tau)$ recovers the convolution integral

$$x(t) = \int_0^t F(\tau)g(t-\tau)\,d\tau.$$

General use of the convolution integral:

Derive the response of an undamped SDOF oscillator to the following triangular pulse using the convolution technique. Note that this problem can easily be solved using the step and ramp response functions described before.



The system is described by $\ddot{x}(t) + \omega_n^2 x(t) = \dfrac{F(t)}{m}$

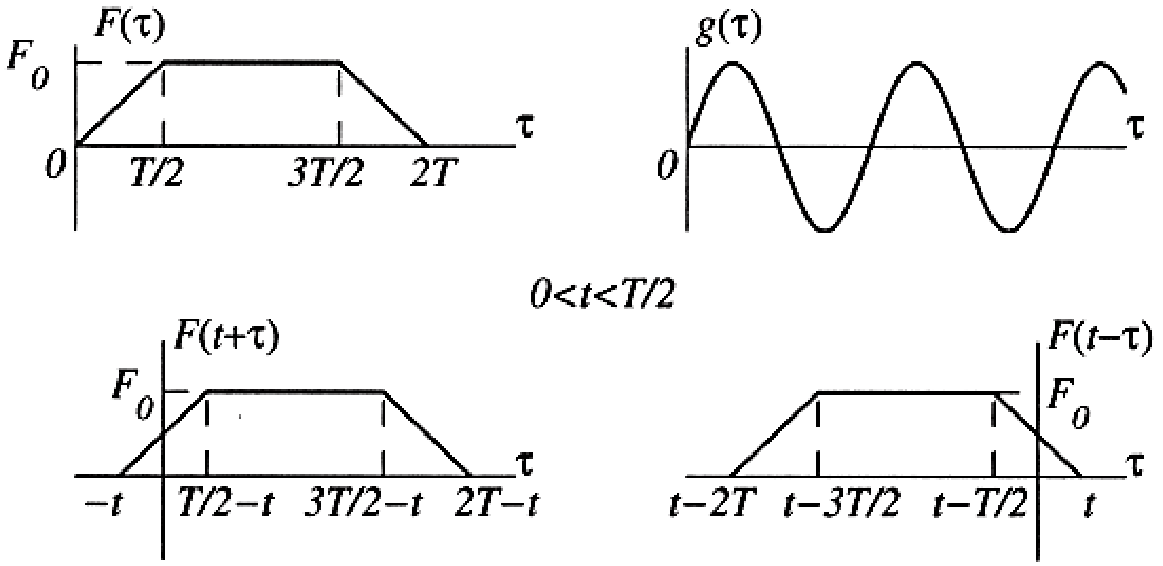
The response is given by the convolution integral in the form $x(t) = \displaystyle\int_0^t F(t-\tau)g(\tau)d\tau$

where

$$F(t) = \begin{cases} \dfrac{2F_0}{T}t, & 0 < t < \dfrac{T}{2} \\ F_0, & \dfrac{T}{2} < t < \dfrac{3T}{2} \\ \dfrac{2F_0}{T}(2T-t), & \dfrac{3T}{2} < t < 2T \\ 0, & t > 2T \end{cases} \quad , g(t) = \frac{1}{m\omega_n}\sin\omega_n t u(t)$$

To evaluate the convolution integral, we must specify the limits of integration. To this end, we consider four cases, $0 < t < T/2$, $T/2 < t < 3T/2$, $3T/2 < t < 2T$ and $t > 2T$, in conjunction with sketches analogous to those shown in Fig. 4.16.

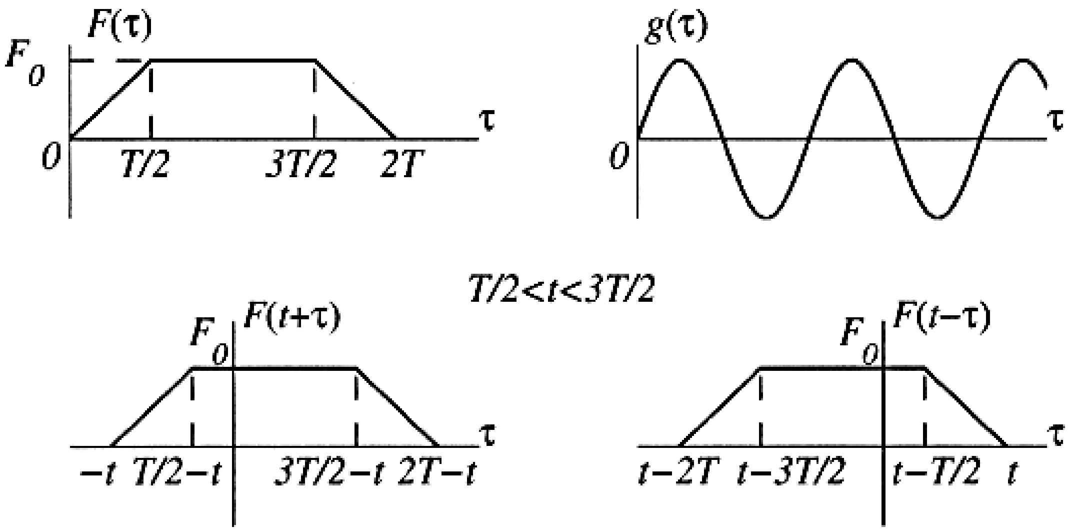
For $0 < t < T/2$, we have



so that there is only one region of integration, $0 < \tau < t$. Hence, the response is

$$x(t) = \int_0^t F(t-\tau)g(\tau)d\tau = \frac{2F_0}{m\omega_n T}\int_0^t (t-\tau)\sin\omega_n\tau d\tau$$
$$= \frac{2F_0}{kT}\left(t - \frac{\sin\omega_n t}{\omega_n}\right), \; 0 < t < \frac{T}{2}$$

For $T/2 < t < 3T/2$, the picture is



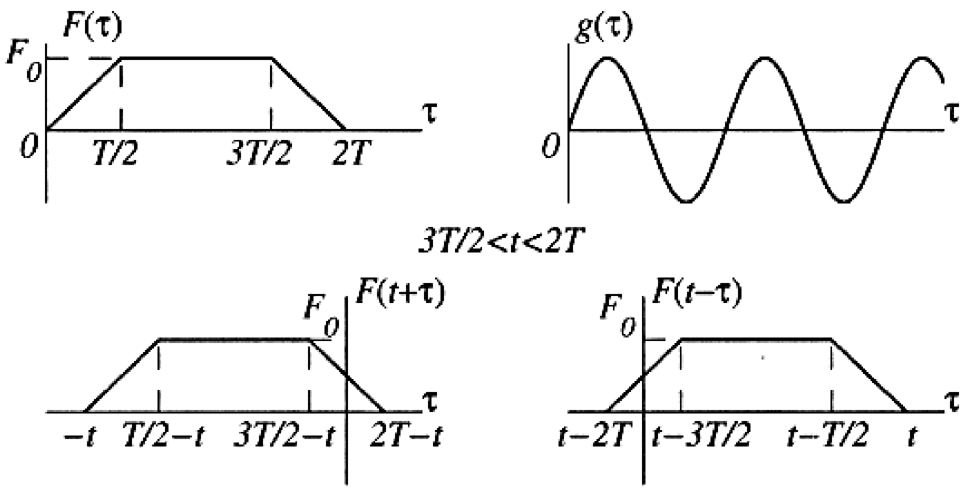
so that there are two regions of integration, $0 < \tau < t - T/2$ and $t - T/2 < \tau < t$. It follows that the response is

$$x(t) = \int_0^t F(t-\tau)g(\tau)d\tau = \frac{F_0}{m\omega_n}\left[\int_0^{t-T/2} \sin\omega_n\tau d\tau + \frac{2}{T}\int_{t-T/2}^{t}(t-\tau)\sin\omega_n\tau d\tau\right]$$

$$= \frac{2F_0}{kT\omega_n}\left[\frac{\omega_n T}{2} + \sin\omega_n\left(t - \frac{T}{2}\right) - \sin\omega_n t\right], \quad \frac{T}{2} < t < \frac{3T}{2}$$

WAYNE STATE UNIVERSITY

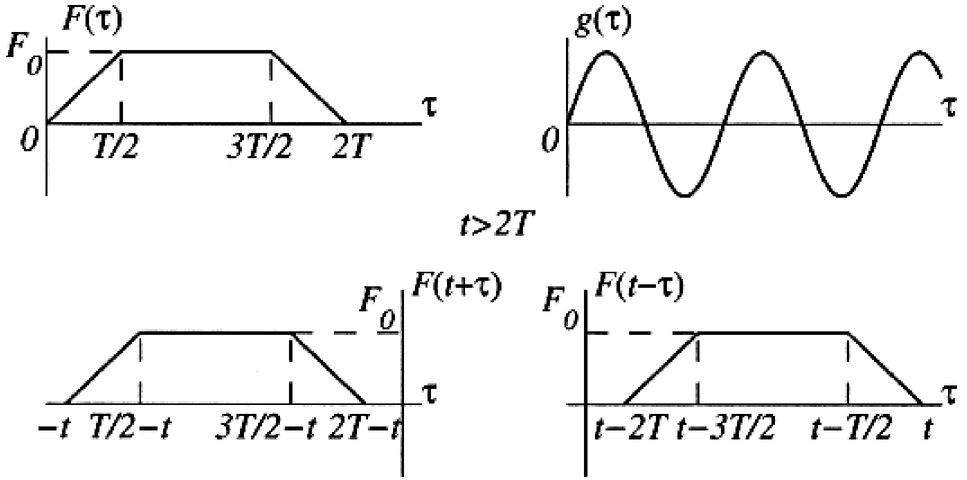
For $3T/2 < t < 2T$, we have



so that now there are three regions of integration, $0 < \tau < t-3T/2$, $t-3T/2 < \tau < t-T/2$ and $t - T/2 < \tau < t$. It follows that the response is

$$
\begin{aligned}
x(t) &= \int_0^t F(t-\tau)g(\tau)d\tau = \frac{F_0}{m\omega_n}\left[\frac{2}{T}\int_0^{t-3T/2}[2T-(t-\tau)]\sin\omega_n\tau d\tau\right. \\
&\quad \left. + \int_{t-3T/2}^{t-T/2}\sin\omega_n\tau d\tau + \frac{2}{T}\int_{t-T/2}^{t}(t-\tau)\sin\omega_n\tau d\tau\right] \\
&= \frac{2F_0}{kT\omega_n}\left[\omega_n(2T-t) + \sin\omega_n\left(t-\frac{T}{2}\right) + \sin\omega_n\left(t-\frac{3T}{2}\right) - \sin\omega_n t\right], \quad \frac{3T}{2} < t < 2T
\end{aligned}
$$

Finally, for $t > 2T$, the limits of integration can be obtained from



so that there are still three regions of integration, but this time they are $t - 2T < \tau < t - 3T/2$, $t - 3T/2 < t - T/2$ and $t - T/2 < \tau < t$. Hence the response is

$$
\begin{aligned}
x(t) = \int_0^t F(t-\tau)g(\tau)d\tau &= \frac{F_0}{m\omega_n}\left[\frac{2}{T}\int_{t-2T}^{t-3T/2}[2T - (t-\tau)]\sin\omega_n\tau d\tau\right. \\
&\left. + \int_{t-3T/2}^{t-T/2}\sin\omega_n\tau d\tau + \frac{2}{T}\int_{t-T/2}^{t}(t-\tau)\sin\omega_n\tau d\tau\right] \\
&= \frac{2F_0}{kT\omega_n}\left[-\sin\omega_n t + \sin\omega_n\left(t - \frac{T}{2}\right) + \sin\omega_n\left(t - \frac{3T}{2}\right) - \sin\omega_n(t-2T)\right], \ t > 2T
\end{aligned}
$$

2.3.3 Response to arbitrary excitation by Fourier integral:

Periodic: $$f(t)=\sum_{-\infty}^{\infty}C_p e^{ip\omega_0 t}=C_0+\sum_{p=1}^{\infty}C_p e^{ip\omega_0 t}+\sum_{p=1}^{\infty}C_{-p} e^{ip\omega_0 t}$$

$$C_p=\frac{1}{T}\int_0^T f(t)e^{-ip\omega_0 t}dt=\frac{1}{T}\int_{-T/2}^{T/2} f(t)e^{-ip\omega_0 t}dt\,,\quad p\in(-\infty,\infty)$$

***f*(*t*) is a real-valued function =>** $C_{-p}=C_p^*$ **(* denotes complex conjugate)**

Non-periodic: **The "period" *T* becomes infinite. Let**

$$p\omega_o=\omega_p,\quad \Delta\omega_p=(p+1)\omega_0-p(\omega_0)=\omega_0=\frac{2\pi}{T}$$

$$f(t)=\sum_{p=-\infty}^{\infty}\frac{1}{T}(TC_p)e^{i\omega_p t}=\sum_{p=-\infty}^{\infty}(TC_p)e^{i\omega_p t}\frac{\Delta\omega_p}{2\pi},\qquad \boxed{f(t)=\frac{1}{2\pi}\int_{-\infty}^{\infty}\hat{f}(\omega)\,e^{i\omega t}d\omega}$$

From, $TC_p=\hat{f}(\omega)\quad\Rightarrow\quad \boxed{\hat{f}(\omega)=\int_{-\infty}^{\infty}f(t)\,e^{-i\omega t}dt}$

These represent a Fourier transform pair

2.3.3 Response to arbitrary excitation by Fourier integral (continue):

Periodic: $\ddot{x} + 2\zeta\omega_n \dot{x} + \omega_n^2 x = \omega_n^2 f(t)\,, \quad x(t) = \sum_{-\infty}^{\infty} C_p G\left(p\omega_0\right) e^{ip\omega_0 t}$

Non-periodic: $\ddot{x}(t) + 2\zeta\omega_n \dot{x} + \omega_n^2 x = \omega_n^2 f(t)\,, \quad x(t) = \dfrac{1}{2\pi}\int_{-\infty}^{\infty} \underbrace{\hat{f}(\omega) G(\omega)}_{X(\omega)} e^{i\omega t}\, d\omega$

Domain	Excitation	Response Expression
Time	$F(t)$	$x(t) = \int_0^t g(t-\tau) F(\tau)\, d\tau$
Frequency	$\hat{f}(\omega)$	$X(\omega) = \hat{f}(\omega) G(\omega)$

$$G(\omega) = \int_{-\infty}^{\infty} g(t)\, e^{-i\omega t}\, dt\,, \quad g(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} G(\omega) e^{i\omega t}\, d\omega$$

where, $g(t) = \left[\dfrac{k}{m\omega_d} e^{-\zeta\omega_n t} \sin\omega_d t\right] \boldsymbol{u}(t)$

Frequency-Response Function (Frequency-Transfer Function)

The *transfer function* is a very useful concept in the frequency domain. For linear vibration, this function contains all the response characteristics of the system.

Impulse Response

The impulse-response function of a system can be obtained by taking the inverse Laplace transform of the system transfer function. For example, consider the damped simple oscillator given by the normalized transfer function:

$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (3.62)$$

The characteristic equation of this system is given by

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0 \quad (3.63)$$

The eigenvalues (poles) are given by its roots. Three possible cases exist, as discussed below and also in Chapter 2.

$m\ddot{x} + c\dot{x} + kx = F(t) = kA\,e^{i\omega t} \Rightarrow \ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2 x = A\omega_n^2\,e^{i\omega t}$

Derivations of *H*(*s*): where $2\zeta\omega_n = \frac{c}{m} \Rightarrow \zeta = \frac{c}{2m\omega_n} = \frac{c}{2\sqrt{mk}}$

Laplace transform of the eqn

$s^2 X + 2\zeta\omega_n s X + \omega_n^2 X = \omega_n^2 \mathcal{L}[Ae^{i\omega t}] \Rightarrow \frac{\text{Output}}{\text{input}} = H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$

WAYNE STATE UNIVERSITY

If input is unit impulse, then $X(s) = H(s)$.

The impulse-response functions $h(t)$ corresponding to the three cases are determined by taking the inverse Laplace transform (Table 3.1) of equation (3.62) for $\zeta < 1$, $\zeta > 1$, and $\zeta = 1$, respectively. The following results are obtained:

$$y_{\text{impulse}}(t) = h(t) = \frac{\omega_n}{\sqrt{1-\zeta^2}} \exp(-\zeta\omega_n t)\sin\omega_d t \qquad \text{for } \zeta < 1 \qquad (3.70a)$$

$$y_{\text{impulse}}(t) = h(t) = \frac{ab}{(b-a)}\left[\exp(-at) - \exp(-bt)\right] \qquad \text{for } \zeta > 1 \qquad (3.70b)$$

$$y_{\text{impulse}}(t) = h(t) = \omega_n^2 t \exp(-\omega_n t) \qquad \text{for } \zeta = 1 \qquad (3.70c)$$

Step Response

Unit step function is defined by

$$\mathcal{U}(t) = 1 \qquad \text{for } t > 0$$
$$= 0 \qquad \text{for } t \le 0 \tag{3.71}$$

Unit impulse function $\delta(t)$ can be interpreted as the time derivative of $\mathcal{U}(t)$; thus,

$$\delta(t) = \frac{d\mathcal{U}(t)}{dt} \tag{3.72}$$

Note that equation (3.72) reestablishes the fact that for nondimensional $\mathcal{U}(t)$, the dimension of $\delta(t)$ is $(\text{time})^{-1}$. Because $\mathcal{L}\mathcal{U}(t) = 1/s$ (see Table 3.1), the unit step response of the dynamic system (3.62) can be obtained by taking the inverse Laplace transform of

$$Y_{\text{step}}(s) = \frac{1}{s}\frac{\omega_n^2}{\left(s^2 + 2\zeta\omega_n s + \omega_n^2\right)} \tag{3.73}$$

$$y_{\text{step}}(t) = 1 - \frac{1}{\sqrt{1-\zeta^2}}\exp(-\zeta\omega_n t)\sin(\omega_d t + \phi) \qquad \text{for } \zeta < 1 \tag{3.74a}$$

$$y_{\text{step}} = 1 - \frac{1}{(b-a)}\left[b\exp(-at) - a\exp(-bt)\right] \qquad \text{for } \zeta > 1 \tag{3.74b}$$

$$y_{\text{step}} = 1 - (\omega_n t + 1)\exp(-\omega_n t) \qquad \text{for } \zeta = 1 \tag{3.74c}$$

3.2.3 Transfer Function Matrix

Consider again the state-space model of a linear dynamic system as discussed in Chapter 2 and Appendix A. It is given by

$\vec{x} = \begin{Bmatrix} z \\ \dot{z} \end{Bmatrix}$, $A = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{bmatrix}$

$B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $u = f(t)$

$m\ddot{z} + c\dot{z} + kz = f(t) \Rightarrow \frac{d}{dt}\begin{Bmatrix} z \\ \dot{z} \end{Bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{bmatrix}\begin{Bmatrix} z \\ \dot{z} \end{Bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} f$

$$\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{u} \tag{3.76a}$$

$$\boldsymbol{y} = \boldsymbol{C}\boldsymbol{x} + \boldsymbol{D}\boldsymbol{u} \tag{3.76b}$$

If $y = z$, then $C = [1 \ 0]$, $D = 0$

To obtain this relation, Laplace transform the equations (3.76a) and (3.76b) and use zero initial conditions for $\boldsymbol{x}$; thus,

If $y = \dot{z}$, then $C = [0 \ 1]$, $D = 0$

$(sI - A)X(s) = BU(s)$

$\Rightarrow X(s) = (sI - A)^{-1} B U(s)$

$\leftarrow$

$$s\boldsymbol{X}(s) = \boldsymbol{A}\boldsymbol{X}(s) + \boldsymbol{B}\boldsymbol{U}(s) \tag{3.77a}$$

If $y = \begin{Bmatrix} z \\ \dot{z} \end{Bmatrix}$, then

$C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $D = 0$

$$\boldsymbol{Y}(s) = \boldsymbol{C}\boldsymbol{X}(s) + \boldsymbol{D}\boldsymbol{U}(s) \tag{3.77b}$$

Note:

$\det\left[(sI - A)\big|_{s = j\omega_n} \right] = 0$

$$\boldsymbol{Y}(s) = \left[\boldsymbol{C}(s\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{B} + \boldsymbol{D}\right]\boldsymbol{U}(s) \tag{3.79a}$$

or

$$\boldsymbol{Y}(s) = \boldsymbol{G}(s)\boldsymbol{U}(s) \tag{3.79b}$$

The transfer-function matrix $\boldsymbol{G}(s)$ is an $m \times r$ matrix given by

$$\boldsymbol{G}(s) = \boldsymbol{C}(s\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{B} + \boldsymbol{D} \tag{3.80}$$

Computationally speaking, this expression is impractical to evaluate. Later on, we will learn another expression in experimental modal analysis.

Example 3.4

a. Briefly explain an approach that one could use to measure the resonant frequency of a mechanical system. Do you expect this measured frequency to depend on whether displacement, velocity or acceleration is used as the response variable? Justify your answer.

b. A vibration test setup is schematically shown in Figure 3.7.

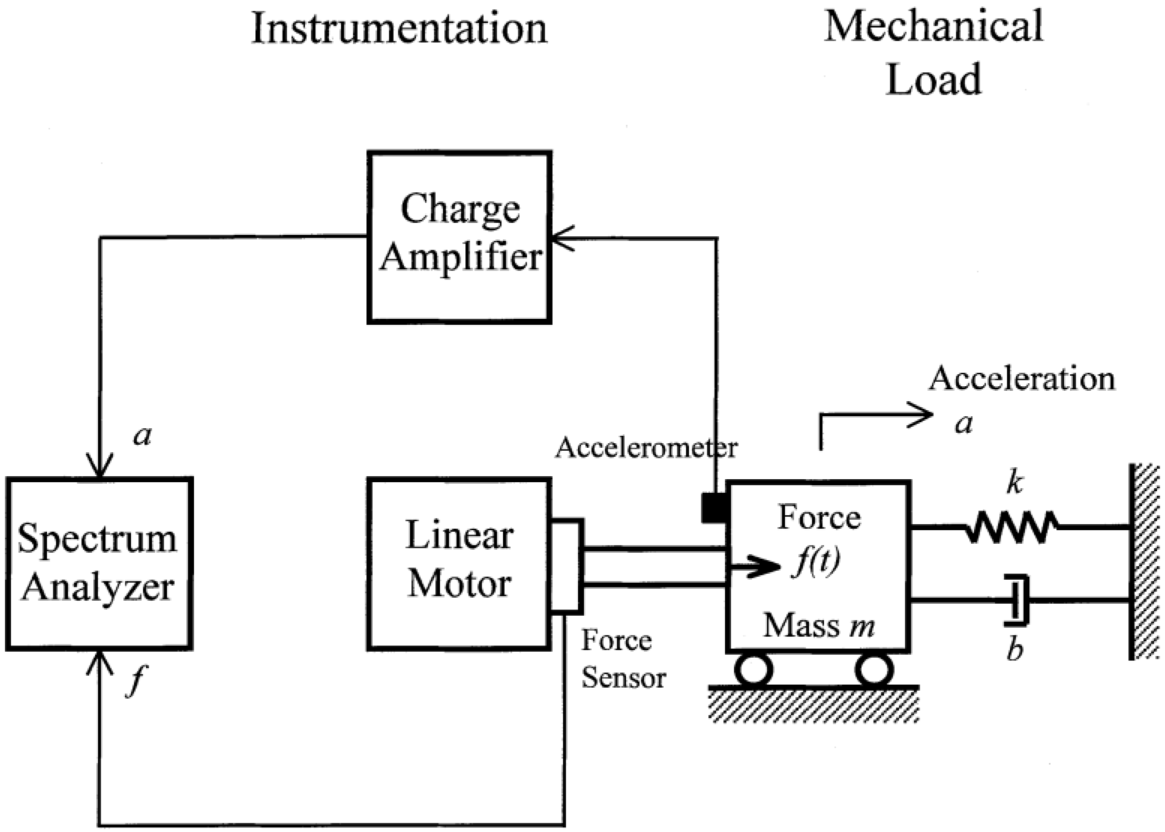


FIGURE 3.7 Measurement of the acceleration spectrum of a mechanical system.

SOLUTION

(a) For a single-degree-of-freedom (dof) system, apply a sinusoidal forcing excitation at the dof and measure the displacement response at the same location. Vary the excitation frequency ω in small steps and, for each frequency at steady state, determine the amplitude ratio of the [displacement response/forcing excitation]. The peak amplitude ratio will correspond to the resonance. For a multi-dof system, several tests may be needed, with excitations applied at different locations of freedom and the response measured at various locations as well. (See Chapters 10 and 11). In the frequency domain:

$$|\text{Velocity response spectrum}| = \omega \times |\text{Displacement response spectrum}|$$

$$|\text{Acceleration response spectrum}| = \omega \times |\text{Velocity response spectrum}|$$

It follows that the shape of the frequency response function will depend on whether the displacement, velocity, or acceleration is used as the response variable. Hence, it is likely that the frequency at which the peak amplification occurs (i.e., resonance) will also depend on the type of response variable that is used.

(b) A free-body diagram of the mass element is shown in Figure 3.8.

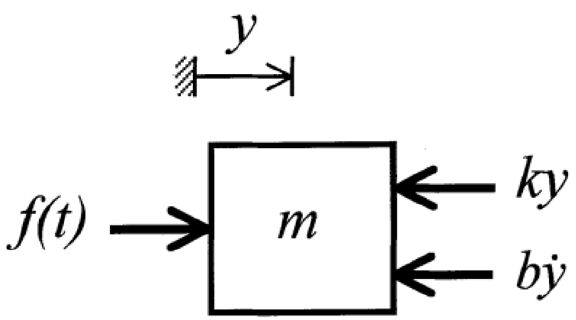


FIGURE 3.8 Free-body diagram.

The displacement transfer function is

$$\frac{y}{f} = \frac{1}{\left(ms^2 + bs + k\right)} \qquad (iii)$$

$$\frac{a}{f} = \frac{s^2}{\left(ms^2 + bs + k\right)} = G(s) \qquad (iv)$$

In the frequency domain, the corresponding frequency response function is obtained by substituting $j\omega$ for s. Hence,

$$G(j\omega) = \frac{-\omega^2}{\left(-m\omega^2 + bj\omega + k\right)} \qquad (v)$$

$$G(j\omega) = \frac{-\omega^2/m}{\left(\omega_n^2 - \omega^2 + 2j\zeta\omega_n\omega\right)} = \frac{-r^2/m}{1 - r^2 + 2j\zeta r} \qquad (vi)$$

where $r = \dfrac{\omega}{\omega_n}$. The magnitude of $G(j\omega)$ gives the amplification of the acceleration signal with respect to the forcing excitation:

$$\left|G(j\omega)\right| = \frac{r^2/m}{\sqrt{\left(1 - r^2\right)^2 + (2\zeta r)^2}} \qquad (vii)$$

Peak value occurs when:

$$r = \frac{1}{\sqrt{1 - 2\zeta^2}} \qquad (ix)$$

Hence,

$$|G|_{\text{peak}} = \frac{1}{2m\zeta\sqrt{1 - \zeta^2}} \qquad (x)$$